MRPL — Mangalore Refinery and Petrochemicals Ltd.

PUMP P-101 MAINTENANCE REPORT — Q2 2025

Equipment ID:	Pump P-101
Equipment Name:	Crude Oil Feed Pump — Unit 3
Location:	Process Area 3, Block B
Report Date:	15-June-2025
Prepared By:	Rajan K., Senior Maintenance Engineer
Approved By:	Suresh N., Maintenance Manager
Classification:	Internal — Restricted

1. Executive Summary

This report covers the quarterly preventive maintenance and inspection of Centrifugal Pump P-101 located in Process Area 3, Unit 3 of the MRPL refinery. The pump was taken offline on 10-June-2025 for scheduled maintenance. Inspection revealed significant wear on the mechanical seal and early-stage bearing degradation. Both components were replaced. The pump was returned to service on 13-June-2025 following successful commissioning tests.

2. Equipment Specifications

Parameter	Value	Unit
Pump Type	Centrifugal, Single-Stage	—
Manufacturer	Grundfos / KSB	—
Model	NK 100-250/260 A2-F-A-E-BQQE	—
Flow Rate (Design)	320	m³/h
Head (Design)	85	m
Speed	1480	RPM
Motor Power	110	kW
Operating Temp.	65–85	°C
Fluid Handled	Crude Oil (SG 0.87)	—
Suction Pressure	1.2	bar(g)
Discharge Pressure	9.8	bar(g)

3. Inspection Findings

3.1 Mechanical Seal

The mechanical seal (John Crane Type 1 equivalent, 50mm shaft) exhibited heavy scoring on the stationary face. Leakage was observed at approximately 12 drops per minute at the seal flush connection, significantly above the acceptable threshold of 3 drops per minute. The seal faces showed signs of dry-running, likely caused by intermittent loss of flush fluid pressure during the upstream vessel low-level event recorded on 02-April-2025.

3.2 Bearings — Drive End (DE)

De-Energy and Lock Out / Tag Out (LOTO) was applied per SOP-PM-007. Bearing SKF 6316 (DE) was inspected. Vibration readings taken prior to shutdown: radial 4.2 mm/s RMS at DE, 2.1 mm/s at NDE. ISO 10816-3 alarm threshold for this pump class is 4.5 mm/s; the DE reading was approaching alarm. Bearing disassembly revealed slight pitting on the outer race. Grease analysis indicated contamination with particulate matter consistent with seal material debris. Bearing replaced with OEM equivalent.

3.3 Bearings — Non-Drive End (NDE)

NDE bearing (SKF 6313) was within acceptable condition. Grease was refreshed. No replacement required.

3.4 Impeller

Impeller (316 SS, semi-open, 5-vane) was removed and inspected. Minor erosion pitting observed on leading edges — within acceptable wear limits. Impeller clearance reset to 0.3 mm per OEM specification. No replacement necessary at this interval.

3.5 Shaft Alignment

Post-reassembly laser alignment check performed using Pruftechnik Rotalign Ultra. Final alignment values: Angular offset 0.02 mm/100mm (limit 0.05), Parallel offset 0.03 mm (limit 0.10). Within tolerance.

4. Bearing Temperature History (Pre-Maintenance)

The following bearing temperatures were recorded via the plant DCS over the 90 days prior to this maintenance event. Elevated temperatures at the DE bearing were the primary trigger for scheduling maintenance ahead of the normal 6-month interval.

Date	DE Bearing Temp (°C)	NDE Bearing Temp (°C)	Alarm Limit (°C)
01-Mar-2025	72	68	85
15-Mar-2025	74	69	85
01-Apr-2025	77	70	85
15-Apr-2025	79	71	85
01-May-2025	81	70	85
15-May-2025	84	71	85
28-May-2025	**87 ■**	71	85
05-Jun-2025	**89 ■**	72	85
10-Jun-2025	Shutdown	Shutdown	—

■ Bearing temperature exceeded alarm setpoint (85°C) on 28-May-2025. Maintenance team was notified. Visual inspection on 29-May confirmed seal leakage. Decision taken to advance scheduled maintenance to 10-June-2025.

5. Work Performed

- Complete pump disassembly per Equipment Manual EM-P101-2019, Section 7.
- Mechanical seal replacement: John Crane Type 1, 50mm, Part No. JC-50-316-TC.
- DE bearing replacement: SKF 6316/C3, lubricated with Shell Gadus S3 V220C grease.
- NDE bearing grease refresh: Shell Gadus S3 V220C, 25g applied.
- Impeller clearance reset to 0.3 mm (OEM specification).
- Shaft run-out check: Max 0.025 mm TIR — within limit.
- Casing wear ring clearance check: 0.45 mm — within acceptable limit.
- Coupling element (Rexnord Thomas disc coupling) inspected — no replacement needed.
- Complete reassembly with new gaskets and fastener re-torquing per EM-P101-2019, Table 9.
- Laser alignment: Angular 0.02 mm/100mm, Parallel 0.03 mm — in tolerance.
- Seal flush line flushed and pressure tested at 4 bar.
- Pre-commissioning run (1 hour at 50% flow): Bearing temps DE 68°C, NDE 65°C — acceptable.
- Full flow commissioning (2 hours): DE 71°C, NDE 67°C — stable.

6. Recommendations

1. Investigate the root cause of the upstream vessel low-level event (02-April-2025) that caused seal dry-running. Implement a low-level interlock on the seal flush supply vessel.

2. Reduce bearing temperature alarm setpoint from 85°C to 80°C for Pump P-101 given the observed trend of accelerated bearing degradation.

3. Increase maintenance frequency from 6-monthly to quarterly for P-101 until root cause of seal and bearing degradation is fully resolved.

4. Consider installing online vibration monitoring (accelerometer) on P-101 DE bearing to enable continuous trending without manual readings.

Report prepared in compliance with MRPL Maintenance Procedure MP-001 Rev.4 and ISO 17359 (Condition Monitoring). All work performed under PTW No. PTW-2025-0614-003.